THE PREDEFORMATION CATEGORY OF A FINITE-TYPE POINT OF AN ALGEBRAIC STACK IS A DEFORMATION CATEGORY WITH FINITE-DIMENSIONAL TANGENT AND INFINITESIMAL SPACES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 0DQU: sources/stacks-chow/tags.txt, source lines 17199–17199. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07T2: sources/stacks-chow/tags.txt, source lines 9922–9922. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07WU: sources/stacks-chow/tags.txt, source lines 10054–10054. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07WQ: sources/stacks-chow/tags.txt, source lines 10050–10050. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07X1: sources/stacks-chow/tags.txt, source lines 10061–10061. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0DQU let $x_0 : \mathrm{Spec}(k) \to \mathcal{X}$ be a morphism, where k is a finite type field over S. Then $\mathcal{F}_{\mathcal{X},k,x_0}$ is a deformation category and $T\mathcal{F}_{\mathcal{X},k,x_0}$ and $Inf(\mathcal{F}_{\mathcal{X},k,x_0})$ are finite dimensional k-vector spaces.*

Proof. Choose an affine open $\mathrm{Spec}(\Lambda) \subset S$ such that $\mathrm{Spec}(k) \to S$ factors through it. By Artin's Axioms, Section 07T2 we obtain a predeformation category $\mathcal{F}_{\mathcal{X},k,x_0}$ over the category $\mathcal{C}_\Lambda$. (As pointed out in locus citatus this category only depends on the morphism $\mathrm{Spec}(k) \to S$ and not on the choice of Λ.) By Artin's Axioms, Lemmas 07WU and 07WQ $\mathcal{F}_{\mathcal{X},k,x_0}$ is actually a deformation category. By Artin's

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Axioms, Lemma 07X1 we find that $T\mathcal{F}_{\mathcal{X},k,x_0}$ and $\mathrm{Inf}(\mathcal{F}_{\mathcal{X},k,x_0})$ are finite dimensional k-vector spaces. $\square$